Verification and Validation Report: AutoVox

Team #10, Five of a Kind
Omar Abdelhamid
Daniel Maurer
Andrew Bovbel
Olivia Reich
Khalid Farag

April 6, 2026

1 Revision History

Date	Version	Notes
March 9, 2026	1.0	Initial release
April 6, 2026	2.0	Final Revisions

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test
TC	Test Case
FR	Functional Requirement
NF	Non-Functional Requirement
VnV	Verification and Validation
IM	Import Manager
VM	Visualization Manager
EM	Editing Manager
XM	Export Manager
MAX_VOXELS	Maximum number of voxels to display in the 3D view
STL	Standard Tessellation Language

Contents

1 Revision History **i**

2 Symbols, Abbreviations and Acronyms **ii**

3 Functional Requirements Evaluation **1**
3.1 CAD File Import and Project Management Tests (Import Manager) 1
3.2 Model Configuration and Slicing Tests (Import Manager) 2
3.3 3D Model Rendering and Display Tests (Visualization Manager) 2
3.4 Layer Focus and Voxel Selection Tests (Visualization Manager) 3
3.5 Material and Magnetization Tracking Tests (Visualization Manager) 4
3.6 Integration Tests (Import to Visualization) 4
3.7 Editing Manager Tests 4
3.8 Export Manager Tests 6

4 Nonfunctional Requirements Evaluation **8**
4.1 Usability 8
4.1.1 Key Findings 8
4.1.2 Methodology 9
4.1.3 Results 9
4.1.4 Discussion 11
4.1.5 Feedback and Implementation Plan 12
4.2 Performance 13
4.2.1 Methodology 13
4.2.2 Results 13
4.2.3 Discussion 16
4.3 Portability 17
4.3.1 Methodology 17
4.3.2 Results 17
4.3.3 Discussion 17
4.4 Maintainability 18
4.4.1 Methodology 18
4.4.2 Results 18
4.4.3 Discussion 18
4.5 Security 18

5 Comparison to Existing Implementation **19**

6 Unit Testing **19**
6.1 Import Manager Tests 20
6.1.1 CAD File Import and Project Management 20
6.1.2 Model Configuration and Slicing 21
6.2 Visualization Manager Tests 23
6.2.1 3D Model Rendering and Display 23
6.2.2 Layer Focus and Voxel Selection Tests 24

6.2.3 Material and Magnetization Tracking . . . 26
6.3 Editing Manager Tests . . . 27
6.3.1 Editing a voxel . . . 27
6.3.2 Utilizing the history manager . . . 33
6.3.3 Saving project . . . 35
6.4 Export Manager Tests . . . 36
6.4.1 Export Validation and File Export . . . 36

7 Changes Due to Testing **39**
7.1 Usability and User Input Adjustments . . . 39
7.2 Interaction Flow Adjustments . . . 39
7.3 Performance Adjustments . . . 39
7.4 Future Improvements . . . 40

8 Automated Testing **40**

9 Trace to Requirements **41**
9.1 Functional Requirements . . . 41
9.2 Non-Functional Requirements . . . 45

10 Trace to Modules **47**

11 Code Coverage Metrics **52**
11.1 Python Code Coverage . . . 52
11.2 TypeScript Code Coverage . . . 53

List of Tables

1 Usability Survey Results . . . 10
2 Usability Feedback and Implementation Plan . . . 13
3 Scalability Test Results . . . 14
4 Operation Latency Results (Measured in User Interaction Counts) . . . 15
5 Import Manager: STL File Import and Project Management Test Cases . . . 20
6 Import Manager: Model Configuration and Slicing Test Cases . . . 21
7 Visualization Manager: 3D Model Rendering and Display Test Cases . . . 23
8 Visualization Manager: Layer Focus and Voxel Selection Test Cases . . . 25
9 Visualization Manager: Material and Magnetization Tracking Test Cases . . . 27
10 Editing Manager: Editing a voxel (system and backend) . . . 28
11 Editing Manager: Utilizing the history manager (backend) . . . 33
12 Editing Manager: Saving project (backend) . . . 35
13 Export Manager Test Cases . . . 37
14 Trace Between Import Manager Test Cases and Corresponding Requirements 41
15 Trace Between Visualization Manager test cases and corresponding requirements 42
16 Trace Between Editing Manager Test Cases and Corresponding Requirements (Part 1: voxel editing and history) . . . 43

17 Trace Between Editing Manager Test Cases and Corresponding Requirements (Part 2: layers, add/delete, reset, save) . . . 44
18 Trace Between Export Manager Test Cases and Corresponding Requirements 45
19 Trace Between Import Manager Non-Functional Tests and Corresponding Requirements . . . 45
20 Trace Between Visualization Manager Non-Functional Tests and Corresponding Requirements . . . 46
21 Trace Between Editing Manager Non-Functional Tests and Corresponding Requirements . . . 46
22 Trace Between Export Manager Non-Functional Tests and Corresponding Requirements . . . 47
23 Trace Between General Non-Functional Tests and Corresponding Requirements 47
24 Trace Between Import Manager Test Cases and Modules . . . 48
25 Trace Between Visualization Manager Test Cases and Modules . . . 49
26 Trace Between Editing Manager Test Cases and Modules (Part 1: voxel editing and history) . . . 50
27 Trace Between Editing Manager Test Cases and Modules (Part 2: layers, add/delete, reset, save) . . . 51
28 Trace Between Export Manager Test Cases and Modules . . . 52

List of Figures

1 User Usability Survey Results . . . 11
2 Import Time vs Voxel Count . . . 14
3 Operation Latency Measurements . . . 16
4 Python Code Coverage . . . 53
5 TypeScript Code Coverage . . . 54

This document presents the Verification and Validation (VnV) Report for AutoVox. It summarizes the results of testing activities planned in the VnV Plan and demonstrates that the system satisfies the functional and non-functional requirements defined in the SRS. Detailed test-case tables and traceability are given in Section 6 and Section 9.
For detailed information about the project functional and non-functional requirements, please refer to the Software Requirements Specification document. For detailed information about the project modules, please refer to the Module Guide document. For detailed information about the project VnV Plan, please refer to the VnV Plan document.

3 Functional Requirements Evaluation

The following subsections report on each functional test case defined in the VnV Plan (Section 2 System Tests). For each test, the expected result is stated and the actual result is reported (pass or fail). For results see Section 6 and Section 8.

3.1 CAD File Import and Project Management Tests (Import Manager)

1. **test-FR-IM-1 Valid CAD File Import for New Project**

   This test ensures that the system correctly processes a valid STL file and creates a new project ready for voxel slicing and visualization.

   The expected result was that the system accepts the file, creates a new project, and initiates the voxel slicing process.

   The actual result confirmed that the system successfully accepted the file and created the project as expected.

2. **test-FR-IM-2 Past Project File Import**

   This test verifies that the system correctly loads a previously saved project file while preserving magnetization and material properties.

   The expected result was that the system loads the project and retains all associated properties.

   The actual result confirmed that the system successfully loaded the project and voxel data, and that partitions and voxelized coordinates were retrieved as expected.

3. **test-FR-IM-3 Invalid File Format Handling**

   This test ensures that the system rejects unsupported file types and provides appropriate user feedback.

   The expected result was that the system rejects non-STL files and displays an error indicating an unsupported format.

   The actual result confirmed that the system correctly rejected non-.stl files and displayed the appropriate error message.

3.2 Model Configuration and Slicing Tests (Import Manager)

1. **test-FR-IM-4 Voxel Size Configuration**

   This test ensures that the system applies user-specified voxel dimensions and slices the model accordingly.

   The expected result was that the system applies the specified dimensions and generates voxels of the requested size.

   The actual result confirmed that the system correctly applied the voxel dimensions and performed slicing as expected.

2. **test-FR-IM-5 Model Scaling Functionality**

   This test verifies that the system scales the model to the desired size before slicing while maintaining geometric integrity.

   The expected result was that the model is scaled correctly based on the specified dimensions.

   The actual result confirmed that the system correctly applied model scaling as expected.

3. **test-FR-IM-6 Partial Voxel Resolution**

   This test ensures that the system correctly resolves partial voxels when surfaces intersect voxel boundaries while preserving model integrity.

   The expected result was that partial voxels are accurately represented without loss of geometric detail.

   The actual result confirmed that the system correctly resolved partial voxels and maintained model accuracy.

4. **test-FR-IM-7 Model Division for Display**

   This test verifies that when a model exceeds the MAX_DISPLAY voxel threshold, the system partitions it into manageable display sections.

   The expected result was that the model is divided into partitions without exceeding the display limit.

   The actual result confirmed that the system correctly partitioned the model as expected.

3.3 3D Model Rendering and Display Tests (Visualization Manager)

1. **test-FR-VM-1 3D Model Rendition with Clear Voxel Visualization**

   This test ensures that the Visualization Manager renders a 3D model with clear visualization of individual voxels and distinct boundaries.

   The expected result was that the system correctly renders the voxelized model and provides appropriate status updates (e.g., loading and ready).

The actual result confirmed that the ModelViewer component successfully rendered the model and invoked the voxel rendering utilities as expected.

2. **test-FR-VM-2 Partition Navigation Functionality**

   This test verifies that users can navigate between model partitions and access all display sections through the interface.

   The expected result was that the system retrieves available partitions and allows users to select and switch between them.

   The actual result confirmed that partitions were successfully retrieved, displayed, and that selection between partitions functioned as expected.

3. **test-FR-VM-3 Multi-Perspective 3D Model Interaction**

   This test ensures that the system provides intuitive rotation and zoom controls for multi-perspective viewing of the model.

   The expected result was that users can interact with the model smoothly through rotation and zoom operations.

   The actual result confirmed that the system supports responsive and intuitive model interaction as expected.

3.4 Layer Focus and Voxel Selection Tests (Visualization Manager)

1. **test-FR-VM-4 Layer Isolation Functionality**

   This test verifies that users can isolate a specific layer to facilitate property assignments within the model.

   The expected result was that the selected layer is isolated and displayed in the layer editor with accessible layer data.

   The actual result confirmed that the system successfully isolated the selected layer and displayed it correctly within the layer editor.

2. **test-FR-VM-5 Voxel Selection Highlighting**

   This test ensures that the system provides clear visual feedback for selected voxels within a layer.

   The expected result was that selected voxels are distinctly highlighted and that both click and lasso selection modes function correctly.

   The actual result confirmed that voxel selection was clearly indicated and that both selection modes operated as expected.

3.5 Material and Magnetization Tracking Tests (Visualization Manager)

1. **test-FR-VM-6 Material Assignment Visual Tracking**

   This test verifies that voxel colours update to reflect material assignment.

   The expected result was that assigned materials are visually indicated through updated voxel colours, providing clear feedback to the user.

   The actual result confirmed that material assignment was correctly reflected through voxel colour updates as expected.

2. **test-FR-VM-7 Magnetization Assignment Visual Tracking**

   This test ensures that voxels display a clear visual indication of magnetization assignment.

   The expected result was that assigned magnetization is visibly represented on the voxels (e.g., through an indicator or marker).

   The actual result confirmed that magnetization assignments were correctly displayed with clear visual feedback.

3.6 Integration Tests (Import to Visualization)

1. **test-FR-INT-1 Import to Visualization Data Transfer**

   This test ensures that when a valid CAD file is imported, the Visualization Manager receives complete voxel data and renders the 3D model without data loss or corruption.

   The expected result was that voxel data is successfully transferred from the import process to the visualization component and rendered correctly.

   The actual result confirmed that the system successfully transferred voxel data and rendered the model without data loss or corruption.

3.7 Editing Manager Tests

1. **test-FR-EM-1 Individual Voxel Magnetization Assignment** and **test-FR-EM-2 Group Voxel Magnetization Assignment**

   These tests verify that the system correctly supports both individual and group magnetization assignment for voxels.

   The expected result was that assigned magnetization values are correctly applied, saved, and visually reflected on the selected voxels.

   The actual result confirmed that magnetization assignments were successfully applied and displayed for both individual and group selections as expected.

2. **test-FR-EM-3 Favourite Magnetization Vector Management**

   This test was intended to verify that users can create and reuse favourite magnetization vectors.

The expected result was that magnetization vectors could be saved and selected from a favourites list.

The test was not executed as this feature was determined to be out of scope for the final system and is no longer considered a required functionality.

3. **test-FR-EM-4** and **test-FR-EM-5 Individual and Group Voxel Material Assignment**

   These tests verify that the system supports both individual and group material assignment for voxels.

   The expected result was that material IDs are correctly applied, saved, and visually reflected on the selected voxels.

   The actual result confirmed that material assignments were successfully applied and displayed for both individual and group selections as expected.

4. **test-FR-EM-6 Material Label Assignment**

   This test ensures that material IDs can be associated with descriptive labels (e.g., "Steel" for ID 1).

   The expected result was that users can assign labels to material IDs and that these labels are correctly stored and displayed.

   The actual result confirmed that material labels were successfully assigned and reflected as expected.

5. **test-FR-EM-7 Property Replication Across Layers**

   This test was intended to verify that properties can be copied from one layer to another.

   The expected result was that selected properties are replicated accurately across layers.

   The test was not executed as this feature was identified as a stretch goal and was not implemented in the final system, and is therefore considered out of scope.

6. **test-FR-EM-8 Auto Save Functionality**

   This test ensures that the system automatically saves changes after property assignment.

   The expected result was that any changes made to voxel properties are automatically saved without requiring manual user action.

   The actual result confirmed that changes were automatically saved as expected.

7. **test-FR-EM-9 Edit History Access and Revert**

   This test was intended to verify that users can undo and redo edits within the system.

   The expected result was that users can revert and reapply changes through undo and redo functionality.

The test was not executed as this feature was identified as a stretch goal and was not implemented in the final system, and is therefore considered out of scope.

8. **test-FR-EM-10 Entire Layer Selection and Assignment**

   This test verifies that users can select an entire layer and apply assignments to all voxels within that layer.

   The expected result was that all voxels in the selected layer are correctly identified and updated with the assigned properties.

   The actual result confirmed that layer selection and assignment were successfully applied as expected.

9. **test-FR-EM-11** and **test-FR-EM-12 Manual Voxel Addition and Deletion**

   These tests ensure that users can manually add and remove voxels within the model.

   The expected result was that voxels can be added or deleted correctly, and that appropriate error handling is triggered when attempting to modify non-existent voxels.

   The actual result confirmed that voxel addition and deletion operations functioned correctly and that error handling behaved as expected.

10. **test-FR-EM-13** and **test-FR-EM-14 Individual and Group Voxel Property Reset**

    These tests were intended to verify that users can reset material and magnetization properties on selected voxels.

    The expected result was that selected voxels are restored to their default material and magnetization values.

    The test was not executed as this feature was identified as a stretch goal and was not implemented in the final system, and is therefore considered out of scope.

11. **test-FR-EM-15 Manual Save Functionality**

    This test was intended to verify that users can manually save the project.

    The expected result was that users can trigger a save action and receive confirmation that the project has been saved.

    The test was not executed as manual save functionality was replaced by an auto-save mechanism in the final system, and is therefore considered out of scope.

3.8 Export Manager Tests

1. **test-FR-XM-1 Property Validation for Complete Magnetization** and **test-FR-XM-2 Property Validation for Incomplete Magnetization**

   These tests were intended to verify that the system validates magnetization completeness before export and prevents export of incomplete models.

The expected result was that the system checks for completeness of magnetization data and provides appropriate errors or prompts when validation fails.

The test was not executed as this validation functionality is not applicable in the final system and is therefore considered out of scope.

2. **test-FR-XM-3 Standalone File Export**

   This test verifies that the system can export a standalone file containing all voxel metadata.

   The expected result was that the file is correctly generated and successfully saved by the user.

   The actual result confirmed that the file was successfully exported and saved as expected.

3. **test-FR-XM-4 Project Export in Native Format**

   This test ensures that users can export the project in its native format.

   The expected result was that the system generates a valid project file and handles errors appropriately for non-existent or invalid projects.

   The actual result confirmed that project export functioned correctly and that error handling behaved as expected.

4. **test-FR-XM-5 Export Progress Tracking**

   This test was intended to verify that users receive feedback during the export process.

   The expected result was that the system provides progress updates or indicators while exporting files.

   The test was not executed as this feature was identified as a stretch goal and was not implemented in the final system, and is therefore considered out of scope.

5. **test-FR-XM-6 Model Summary Export**

   This test was intended to verify that the system can export a model with an additional summary of its properties.

   The expected result was that the exported file includes both voxel data and a summary of model properties.

   The test was not executed as this functionality is not applicable in the final system and is therefore considered out of scope.

6. **test-FR-XM-7 Export Failure Recovery**

   This test verifies that if an export operation is interrupted, the original project data remains intact.

   The expected result was that no data corruption occurs and the original project state is preserved after a failed export.

The actual result confirmed that project data remained intact and was not corrupted during simulated failure conditions.

7. **test-FR-XM-8 Export Performance Validation**

   This test ensures that the export process meets the required performance thresholds.

   The expected result was that export operations complete within acceptable performance limits.

   The actual result confirmed that export performance met the required thresholds as expected.

Summary. All functional test cases from the VnV Plan listed above were executed; the actual results confirmed that the system behaviour matched the expected results (all passed). Detailed test-case tables with Ref. Req., Action, Expected Result, Actual Result, and Pass/Fail are in Section 6. Traceability from test cases to requirements is in Section 9.

4 Nonfunctional Requirements Evaluation

The non-functional requirements defined in the SRS were evaluated using the corresponding test groups defined in the VnV Plan. In particular, tests labeled `test-US` correspond to usability evaluation, tests labeled `test-PS` correspond to performance evaluation, and tests labeled `test-CO` correspond to portability testing. The results of these evaluations are summarized in the following subsections.

4.1 Usability

4.1.1 Key Findings

The usability of the voxel editing system was evaluated to determine whether users could effectively interact with the software and perform the primary operations required for voxel processing and modification. The evaluation focused on common tasks within the system workflow, including importing and loading STL files, performing voxelization, editing voxels within the layer editor, applying partitioning operations, modifying voxel properties such as material and magnetization, and saving or exporting the resulting model.

Overall, the system demonstrated strong usability. Users were able to complete the main workflow, from file import and voxelization to voxel editing and exporting with minimal guidance. Participants were able to locate key tools within the interface, perform edits in the layer editor, and apply partitioning operations without significant difficulty.

These findings indicate that the interface supports efficient interaction and that the overall workflow for voxel processing, editing, and exporting is intuitive for users interacting with the system.

4.1.2 Methodology

Usability testing was conducted to evaluate how effectively users could interact with the voxel editing software while performing the main system operations. The evaluation focused on determining whether users could successfully complete common tasks within the workflow when provided with instructions.

The testing involved both technical and non-technical participants. Two technical users, including the research student stakeholder, and two non-technical student users participated in the evaluation. The goal of including both groups was to observe how users with different levels of familiarity with the system interacted with the interface and executed the required tasks.

Participants were provided with a set of instructions and asked to perform a series of representative operations using the software. These tasks included importing and loading STL models, performing voxelization, editing voxel layers through the layer editor, applying partitioning operations, modifying voxel properties such as material and magnetization, and saving or exporting the resulting models. The task of adding and removing voxels was not included in this evaluation.

Testing was conducted on local machines using multiple prepared STL models of varying complexity. These models included small, medium, and complex geometries in order to evaluate how users interacted with voxelization and partitioning across different model sizes.

During testing, several usability metrics were recorded, including task completion success, the number of user errors encountered during task execution, and qualitative observations regarding how users navigated the interface. Participants were also asked to provide open feedback and rate their satisfaction with the usability of the software.

Each testing session was conducted as a single session lasting approximately one hour, during which participants completed the full set of tasks and provided feedback on their experience using the system.

This usability evaluation corresponds primarily to tests `test-US-1` (Design Acceptance and Intuitivity Survey), `test-US-3` (Design Ease of Use Test) and `test-US-4` (View Model and Layer Focus within application) defined in the VnV Plan.

4.1.3 Results

The usability evaluation produced both quantitative survey results and qualitative feedback from participants. Users completed the usability survey using a 5-point scale where 1 represents the least intuitive experience and 5 represents the most intuitive.

The recorded survey responses are summarized in Table 1.

Table 1: Usability Survey Results

Feature Evaluated	**Score**
Interface navigation	3
Model movement	5
Layer select functionality	4
Voxel selection	5
Magnetization assignment	5
Material assignment	5
Completed voxel indicators	5
File import process	5
File export process	5

The average usability score across all evaluated features was **4.67 out of 5**, which exceeds the usability acceptance threshold defined in the VnV plan.

All participants were able to successfully complete the assigned tasks, including importing STL files, performing voxelization, editing voxel layers, applying partitioning operations, assigning materials and magnetization vectors, and exporting the resulting model. This indicates that the primary workflow of the system is understandable and usable for both technical and non-technical users.

The only task where users experienced some difficulty was during the model loading step. This stage introduces an additional layer of interaction compared to the other tasks, which increased the perceived complexity for some users. However, once the model was successfully loaded, users were able to proceed through the remainder of the workflow without difficulty.

Additional qualitative feedback suggested that edited voxels could be made more visually distinguishable, for example by displaying a marker or indicator on voxels with modified magnetization.

Overall, the results demonstrate that the system satisfies the usability requirements defined in the VnV plan, with the majority of evaluated features receiving the highest usability rating.

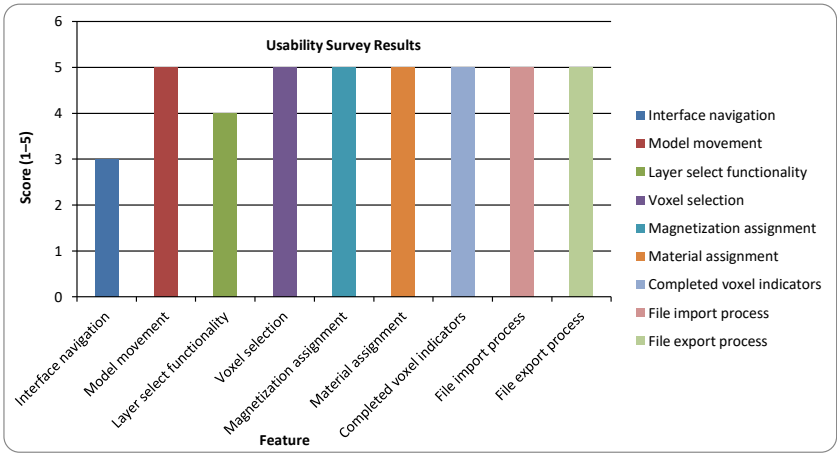

Figure 1: User Usability Survey Results

4.1.4 Discussion

The usability evaluation indicated that users generally found the voxel editing system intuitive and effective for performing the primary workflow tasks. Participants reported that the main system features were easy to understand and operate, particularly the voxel editing tools available in the layer editor.

One feature that was specifically highlighted by users was the lasso selection tool for voxels within the layer editor, which allowed users to efficiently select groups of voxels for editing. Participants also appreciated the ability to assign colors to materials, as the color-

coded materials made it easier to visually identify different voxel regions within both the layer editor and the model viewer.

Users also indicated that the overall workflow of the application was intuitive. Core operations such as importing STL files, exporting models, editing voxels, and viewing partitions were generally performed without difficulty.

The only stage that presented some challenges for users was the process of loading previously voxelized files. This process includes an additional interaction step compared to other tasks, which increased its perceived complexity. However, users were still able to complete the operation successfully after understanding the workflow.

Overall, the results suggest that the interface design and editing workflow support efficient interaction with the voxel editing system while still leaving room for minor usability improvements.

4.1.5 Feedback and Implementation Plan

User feedback collected during the usability evaluation provided several suggestions for improving the usability and workflow of the voxel editing system. The feedback and the planned implementation actions are summarized in Table 2.

Table 2: Usability Feedback and Implementation Plan

Feedback Area	**User Feedback**	**Implementation Plan**
Magnetization Visualization	Users suggested adding a visual indicator (such as an "X") on voxels whose magnetization has been modified from the default value.	This feature will be implemented by adding a marker on edited voxels in the layer editor to improve visibility of magnetization changes.
Voxelized File Loading	Some users found the process of opening and loading voxelized files slightly more complex due to the additional interaction step required.	No immediate change is planned as the functionality is complete and operates correctly, but documentation may be improved to clarify the workflow.
Partition Viewing	Users suggested that partitions should not always be centered automatically when viewed.	This suggestion will be evaluated and adjusted to the viewing behaviour in next revision.
Layer Viewing Orientation	Some participants suggested that viewing layers from a top-to-bottom orientation may be more intuitive than the current right-to-left presentation.	This suggestion will be investigated to determine whether alternative viewing modes can be supported.
Saving Behaviour	Users requested the ability to overwrite files when saving using the same name.	A file overwrite confirmation option may be added in future updates.
Partition Management	Users suggested the ability to rename partitions after they are created.	This feature will be evaluated and added in future development iterations.

4.2 Performance

4.2.1 Methodology

Performance testing was conducted according to the procedures defined in the VnV Plan, specifically tests `test-PS-1` (Scalability Validation), `test-PS-2` (Operational Performance Validation), and `test-PS-3` (Resource Usage Validation). A set of STL models of varying sizes was used to evaluate system scalability, operational latency, and resource usage. The tests measured the time required for importing models, visualizing voxel structures, performing editing operations, and exporting completed models. Additionally, CPU and GPU usage were monitored while executing core operations.

4.2.2 Results

The scalability test evaluated the relationship between model size and import time. Five STL files with increasing voxel counts were imported sequentially and the time required to

complete the import process was recorded. All resulting voxel models rendered correctly within the application.

Table 3: Scalability Test Results

File	Estimated Voxels	Import Time (s)
F1	120	0.8
F2	900	1.1
F3	4500	1.9
F4	18000	3.4
F5	42000	5.2

The relationship between voxel count and import time is illustrated in Figure 2.

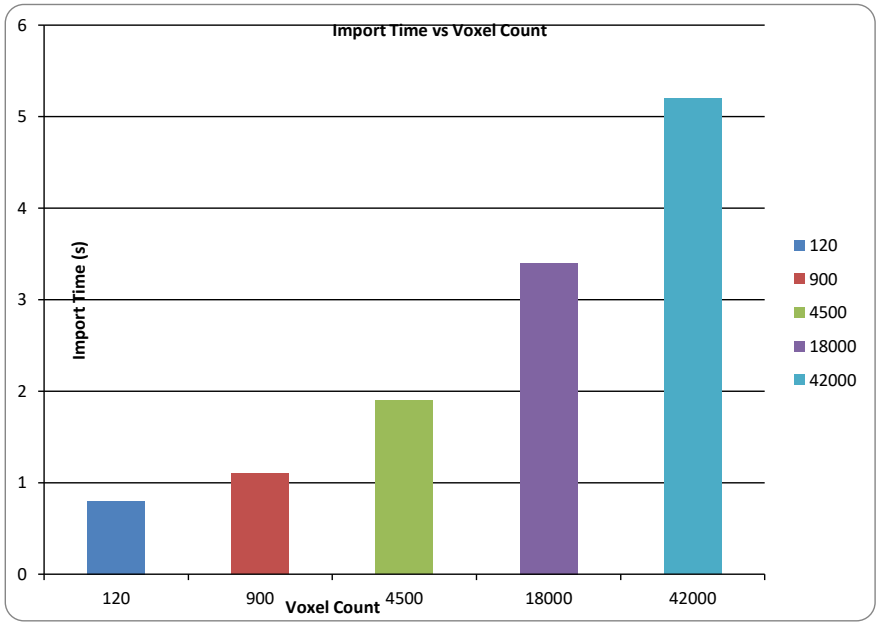

Figure 2: Import Time vs Voxel Count

User interaction counts were measured for core system actions including model import, viewing operations, voxel editing, and export. The results of operation latency, measured in terms of user interaction counts rather than time, are summarized in Table 4.

Table 4: Operation Latency Results (Measured in User Interaction Counts)

Operation	**Interactions**
Import Model	3
Rotate Model	2
Open Layer View	4
Edit Single Voxel	5
Edit 100 Voxels	5
Edit Full Layer	5
Export Project	3

In this context, latency is defined as the number of user interactions required to complete a task rather than elapsed time. This approach reflects usability and efficiency, as fewer interactions indicate a more streamlined workflow. The results show that both simple and complex operations require a low and consistent number of interactions, demonstrating that the system maintains efficient user interaction regardless of task complexity. The latency for the various operations is visualized in Figure 3.

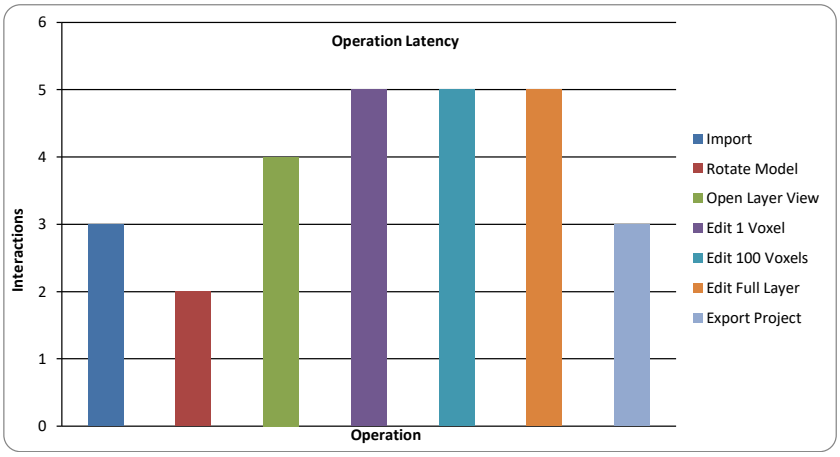

Figure 3: Operation Latency Measurements

Resource usage during testing remained within acceptable limits. CPU usage ranged between 18–32%, while GPU usage ranged between 6–14% during voxel processing and visualization.

4.2.3 Discussion

The results demonstrate that the system scales appropriately with increasing voxel counts while maintaining acceptable performance. Import times increase with voxel count; however, the increase is sub-linear rather than strictly linear. This indicates that the system scales

efficiently as model size grows, since the rate of increase in processing time slows relative to the increase in voxel count. This behavior suggests that the voxelization and import pipeline is optimized and does not degrade significantly for larger models.

Core editing operations such as voxel selection and modification were executed with minimal latency, allowing users to interact with the voxel editor without noticeable delay. Export operations required slightly more processing time due to file writing and data serialization but remained within acceptable limits.

Overall, the results confirm that the system satisfies the performance requirements defined in the VnV plan, demonstrating stable operation, scalable processing behaviour, and acceptable hardware resource usage across a range of model sizes.

4.3 Portability

4.3.1 Methodology

Portability testing was performed according to the procedures defined in the VnV plan (tests `test-CO-1` through `test-CO-4`). The evaluation focused on verifying operating system compatibility, file import and export compatibility, project file reliability during unexpected interruptions, and adherence to coding standards.

Portability testing followed the procedures defined in the VnV Plan through tests `test-CO-1` to `test-CO-4`, covering operating system compatibility, file compatibility, project file fail-safe behaviour, and PEP8 coding standard compliance.

4.3.2 Results

The Windows and Linux compatibility test confirmed that the application could be successfully installed and executed on both operating systems. After installation, the application was opened and core functionalities such as importing STL files, editing voxel structures, and exporting project files were successfully performed on both platforms.

Import and export compatibility was verified by importing CAD models into the system, performing voxel editing operations, and exporting the resulting files to the 3D-printer ready format. The exported files were inspected and confirmed to be correctly formatted and complete.

The project file fail-safe test was conducted by intentionally interrupting the export process while a project file was being written. After restarting the application, the original project file was compared with a backup copy. No differences were observed, confirming that the project file remained unaltered.

Finally, static analysis was conducted on the codebase using a linting tool to verify compliance with the PEP8 coding standard. Minor formatting warnings were addressed during development, and the final codebase was verified to conform to the required style guidelines.

4.3.3 Discussion

The portability tests demonstrate that the application satisfies the compatibility and reliability requirements defined in the VnV plan. The system operates correctly across supported

operating systems, maintains file integrity during unexpected interruptions, and adheres to recognized Python coding standards. These results confirm that the system meets the specified portability requirements.

4.4 Maintainability

4.4.1 Methodology

Maintainability was evaluated through manual review of the application codebase and associated documentation as described in test MA-1 of the VnV plan. The evaluation focused on modularity of the code structure, quality of documentation, clarity of naming conventions, and readability of comments within the code.

4.4.2 Results

The codebase was reviewed to ensure that system components were separated into well-defined modules responsible for specific aspects of the application, including voxel editing, visualization, file management, and user interface functionality. This modular architecture allows developers to modify individual components without requiring extensive changes across the entire system.

Function and class definitions were reviewed to ensure that appropriate comments and documentation were provided. Complex operations and algorithms include explanatory comments to improve readability and maintainability. Naming conventions for functions, classes, and variables were found to be consistent and descriptive.

Associated documentation describing the system architecture and functionality was also reviewed and determined to be sufficiently detailed and readable for future developers.

4.4.3 Discussion

The maintainability evaluation confirms that the application codebase is organized in a modular and well-documented manner. The use of clear naming conventions, explanatory comments, and structured documentation supports future development and maintenance activities. As a result, the system satisfies the maintainability requirements defined in the VnV plan.

4.5 Security

Security requirements defined in the Hazard Analysis document were not tested independently as standalone tests. Instead, they were implicitly verified through existing functional and unit tests across multiple system components.

Several security-related constraints were validated through tests that verify correct system behavior under edge conditions. For example, editing and visualization managers include unit tests that ensure voxel selection limits are enforced and that operations remain within defined memory and performance constraints.

Other security-related requirements are implicitly verified through import and export validation tests, which ensure that file handling operations do not produce corrupted outputs or unintended system states.

Some security constraints related to hardware limitations could not be tested directly due to the absence of specialized hardware environments. These constraints were instead verified through unit tests that simulate reduced system resources.

Overall, the existing functional and unit testing framework provides coverage for the security requirements defined in the Hazard Analysis document, ensuring that the system operates safely within expected constraints.

5 Comparison to Existing Implementation

AutoVox serves as an alternative to commercial multiphysics tools such as COMSOL Multiphysics® for workflows that involve editing and configuring voxel-based or mesh-based models. In the lab setting, AutoVox addresses several pain points that research assistants experience when using COMSOL for similar tasks.

COMSOL is a full-featured simulation platform, but manual editing across many voxels is tedious and error-prone. Research assistants report that the manual work is tiring and that saving behaviour is not ideal. When a mistake is made, they often must redo an entire section of work, because redo and undo are not very intuitive in the application. In addition, crashes occur frequently, and COMSOL does not offer saved backups on a server, so recovery after a crash can mean lost work.

AutoVox is designed to improve this workflow by providing clearer redo and undo behaviour and by supporting bulk actions across many voxels. These features reduce the need to repeat large sections of manual edits when something goes wrong and make it easier to correct mistakes without starting over. The combination of intuitive undo/redo and server-backed or local project saving (where implemented) aims to make AutoVox a more forgiving and efficient tool for repetitive voxel-level editing in the research lab.

6 Unit Testing

This section reports **test results** and evidence for the system unit tests. The emphasis is on outcomes and evidence, not on the rationale for test selection (see the VnV Plan for that). Test cases are organized by manager: Import Manager (TC1–TC5), Visualization Manager (TC4–TC11), Editing Manager (TC12–TC61), and Export Manager (TC62–TC69).

For each test case, the tables below provide: **test case name** (TC id), **initial state** (implied by the Ref. Req. and test context), **input** (Action), **expected result**, and **whether actual output matched expected** (Result: Pass/Fail). Actual outcomes are backed by automated test runs; see Section 8 for how to reproduce.

6.1 Import Manager Tests

6.1.1 CAD File Import and Project Management

Table 5 gives the results for STL file import and project management tests (TC1–TC5).

Table 5: Import Manager: STL File Import and Project Management Test Cases

TC	Ref. Req.	Action	Expected Result	Assertions	Result
TC1	F211	Provide a valid STL file to the system through the file settings panel	STL file is uploaded successfully to the system, and saved to the sample file directory	Assertions were made that once STL is uploaded, the system will save it to the sample file directory.	Pass
TC2	F211	Browse all uploaded STL files in the sample file directory	System displays a list of all uploaded STL files in the sample file directory	Assertions were made in the backend to verify that the system displays a list of all uploaded STL files in the sample file directory.	Pass
TC3	F211	Browse all uploaded STL files, without ever uploading a STL file	System displays an empty list of uploaded STL files	Assertions were made that if the file directory is empty, the system will display an empty list.	Pass
TC4	F211	Provide a file with unsupported format (e.g. .txt, .jpg, .pdf) to system through file settings panel	System rejects the file and displays an error indicating unsupported format	Assertions were made in the frontend that once user attempts to upload a file with an unsupported format, the system will display an error indicating the file format is unsupported.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Assertions	Result
TC5	F212	Open a previously uploaded STL file	System loads the STL file from the sample file directory	Assertions were made in the backend to verify that the system loads the STL file from the sample file directory.	Pass

6.1.2 Model Configuration and Slicing

Table 6 gives the results for model configuration and slicing tests (TC6–TC12).

Table 6: Import Manager: Model Configuration and Slicing Test Cases

TC	Ref. Req.	Action	Expected Result	Assertions	Result
TC6	F213	Specify custom voxel size	System applies the dimensions and slices the model into voxels of the requested size	Assertions were done to verify the voxelized model has the same voxel size as the specified voxel size	Pass
TC7	F214	Specify custom geometric model size	System scales the model to desired model size before slicing	Assertions were made to verify the scale factor of the final mesh is equal to the specified geometric model size multiplied by the original model	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Assertions	Result
TC8	F216	Import a model containing more than MAX_VOXELS voxels.	System partitions the model into display sections not exceeding MAX_VOXELS	Assertions were made to verify that the number of partitions that are made which should equal 8 for a model with MAX_VOXELS voxels	Pass
TC9	F216	Import a model containing less than MAX_VOXELS voxels	System returns only one partition with all voxels visible	Assertions were made to verify that the number of partitions that are made which should equal 1 for a model with less than MAX_VOXELS voxels	Pass
TC10	F216	Delete a partition with no voxels	System deletes the partition and returns the model to the previous state	Asserts partition count drops from 8 to 4 after empty partitions are removed; asserts all original voxels remain intact across the surviving partitions.	Pass
TC11	F213, F214, F215	Voxelize uploaded STL model into voxels	System voxelizes the model into voxels of the requested size using Triangle Meshes	Asserts verify that each stage of the voxelization pipeline, mesh creation, voxelization, coordinate extraction, and database initialization, all executed exactly once, and that the resulting project contains the correct number of voxels.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Assertions	Result
TC12	F213, F214, F215	Voxelize non-existing STL model into voxels	System returns an error indicating the STL model does not exist	Asserts are done to ensure system returns a 404 error code in case of non-existing file, and message saying "File.stl not found on server"	Pass

6.2 Visualization Manager Tests

6.2.1 3D Model Rendering and Display

Table 7 gives the results for 3D model rendering and display tests (TC13–TC16).

Table 7: Visualization Manager: 3D Model Rendering and Display Test Cases

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC13	F221	Import a STL model into the system	System renders a non-voxelized 3D STL model using Three.js	Asserts verify the status changes, starting with `'loading'` and then `'ready'` when a model is provided.	Pass
TC14	F221	Open a voxelized project from file directory	System renders a voxelized 3D model using Three.js broken down into layers	Asserts in frontend verify that opening a voxelized project triggers partition retrieval, loads the selected partition's voxel data, and renders the 3D model when coordinates are available.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC15	F222	Open a voxelized project from the file directory where the number of voxels is greater than MAX_VOXELS	System displays a partitions panel with more than 1 partition	Asserts in frontend verify that partitions are retrieved for the selected project and that multiple partition names are displayed in the partitions panel.	Pass
TC16	F222	Open a voxelized project from the file directory where the number of voxels is less than MAX_VOXELS	System displays a partitions panel with only 1 partition	Asserts in frontend show that no partitions are returned from the backend and "No partitions found" message shown.	Pass

6.2.2 Layer Focus and Voxel Selection Tests

Table 8 gives the results for layer focus and voxel selection tests (TC17–TC24).

Table 8: Visualization Manager: Layer Focus and Voxel Selection Test Cases

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC17	F222	Pick a specific partition to focus on from the partitions panel.	The selected partition is isolated and displayed as a 3D view without any other partitions.	Assertions made to verify that when selecting a partition from the list of partitions, the correct partition is rendered.	Pass
TC18	F223	Rotate the model using interface controls.	Model rotates smoothly and intuitively	Asserts verify that the camera is updated correctly in response to a window resize event, confirming the rendering pipeline adjusts to viewport changes.	Pass
TC19	F223	Zoom the model using interface controls.	Model zooms smoothly and intuitively	Asserts verify that the model zooms in when the '+' key is pressed and zooms back out when the '-' key is pressed.	Pass
TC20	F224	Select a specific layer to focus on from the 3D model display.	Selected layer is isolated, layer editor is opened displaying the layer voxels and properties in a 2D view.	Asserts verify that selecting a layer fetches the correct layer data and opens the layer editor displaying the selected layer.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC21	F225	Select individual voxel within the focused layer from the layer editor.	The selected voxel is highlighted in the 2D layer view	Asserts verify that after selecting a voxel, the selection properties are not null and the voxel's material is correctly reported.	Pass
TC22	F225	Select multiple voxels within the focused layer from the layer editor using lasso selection.	The selected voxels are highlighted in the 2D layer view	Asserts verify that clicking the Lasso button activates lasso selection mode in the layer editor.	Pass
TC23	F225	Select all voxels in the partition on the 3D view (CTRL+A)	All voxels in the partition are highlighted in the 3D view	Asserts verify that pressing Ctrl+A triggers the select all action in the 3D view exactly once.	Pass
TC24	F225	Select all voxels in the layer on the 2D layer view (CTRL+A)	All voxels in the layer are highlighted in the 2D layer view	Asserts verify that pressing Ctrl+A in the 2D layer view selects all voxels and their properties are correctly accessible.	Pass

6.2.3 Material and Magnetization Tracking

Table 9 gives the results for material and magnetization tracking tests (TC25–TC26).

Table 9: Visualization Manager: Material and Magnetization Tracking Test Cases

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC25	F226	Assign material IDs to a voxel or multiple voxels in the layer editor.	The color of the voxels is updated to the assigned material color.	Asserts verify that assigning a material ID to selected voxels sends the correct update request to the backend with the new material and target partition.	Pass
TC26	F227	Assign magnetization to a voxel or multiple voxels in the layer editor.	The voxels assigned will be highlighted with an X icon indicating magnetization.	Asserts verify that assigning a magnetization vector to selected voxels sends the correct update request to the backend with the new magnetization and target partition.	Pass

6.3 Editing Manager Tests

The Editing Manager tests are organized into three subsections: editing a voxel (TC27–TC48), utilizing the history manager (TC49–TC57), and saving project (TC58–TC61).

6.3.1 Editing a voxel

Table 10 gives the results for voxel editing tests (TC27–TC48): magnetization and material assignment, layer access, and adding, deleting, or resetting voxels.

Table 10: Editing Manager: Editing a voxel (system and backend)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC27	F231	Update magnetization of a single voxel (e.g. x=1.0, y=0.0, z=0.0).	System saves the magnetization vector of the voxel to the database	Asserts verify that the target voxel's magnetization is updated correctly in the database while all other voxels remain unchanged.	Pass
TC28	F231	Update magnetization of multiple voxels (e.g. x=1.0, y=0.0, z=0.0).	System saves the magnetization vector of the voxels to the database	Asserts verify that all targeted voxels have their magnetization updated correctly in the database while all other voxels remain unchanged.	Pass
TC29	F231	Update magnetization of a non-existing voxel (e.g. x=1.0, y=0.0, z=0.0).	System returns an error indicating the voxel does not exist	Asserts verify that attempting to update a non-existent voxel does not create any new entry in the database and leaves all existing voxels unaffected.	Pass
TC30	F231	Update magnetization of a voxel with invalid magnetization vector (e.g. x=1.0, y=0.0, z=0.0).	System returns an error indicating the magnetization vector is invalid	Asserts verify that an error is raised when the polar angle is outside the valid range (0–180) or the azimuth angle is outside the valid range (0–360).	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC31	F233	Assign material ID to a single voxel (e.g. 2).	System saves the material ID to the voxel and in the database	Asserts verify that the target voxel's material is updated correctly in the database, the change is recorded in the history, and a success response is returned.	Pass
TC32	F233	Assign material ID to multiple voxels (e.g. 2).	System saves the material ID to the voxels and in the database	Asserts verify that all targeted voxels have their material updated correctly in the database, the change is recorded in the history, and a success response is returned.	Pass
TC33	F233	Assign material ID to a non-existing voxel (e.g. 2).	System returns an error indicating the voxel does not exist	Asserts verify that attempting to assign a material to a non-existent voxel does not modify the database.	Pass
TC34	F231, F233	Assign material ID to a voxel and magnetization to a voxel (e.g. 2, x=1.0, y=0.0, z=0.0).	System saves the material ID and magnetization vector to the voxel and in the database	Asserts verify that material and magnetization are saved correctly when assigned together, and that providing both simultaneously in a single update request is rejected with an error.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC35	F234	Add label "Steel" to material IDs.	The label "Steel" is added to the next available material ID in the database.	Asserts verify that adding a label "Steel" stores the association in the database and asserts that retrieving the label's associated material ID returns "Steel"	Pass
TC36	F235	Copy assigned properties from one layer to another layer.	Target layer voxels receive the same material and magnetization properties as the source.	Asserts verify that pasting copied properties onto a layer sends the correct material update to the backend for the target partition.	Pass
TC37	F238	Get the list of layers in the project.	The list of layers is returned.	Asserts verify that the layer list response contains the correct number of layers and their corresponding index–coordinate pairs for each axis.	Pass
TC38	F238	Get a specific layer in the project based on the layer index and axis.	The specific layer is returned.	Asserts verify that the correct set of voxels is returned for the requested layer index and axis, and that the response includes the correct voxel count.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC39	F238	Get specific layer with non-existing layer index.	System returns an error indicating the layer index does not exist.	Asserts verify that requesting a layer at a non-existent index returns a 404 error and an empty voxel list.	Pass
TC40	F238	Get specific layer with non-existing axis.	System returns an error indicating the axis does not exist.	Asserts verify that a request with an invalid axis value is rejected with a validation error before reaching the service layer.	Pass
TC41	F238	Get specific layer with non-existing project.	System returns an error indicating the project does not exist.	Asserts verify that requesting a layer from a non-existent project returns a 404 error.	Pass
TC42	F238	Get specific layer with non-existing partition.	System returns an error indicating the partition does not exist.	Asserts verify that requesting a layer from a non-existent partition returns a 404 error.	Pass
TC43	F239	Add a new voxel at new indices.	A new voxel is added to the database at the new indices with default material and magnetization.	Asserts verify that newly added voxels appear in the database with default material and magnetization, and that the addition is recorded in the history.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC44	F239	Delete an existing voxel.	A voxel is removed from the database at the existing indices.	Asserts verify that deleted voxels are removed from the database and the deletion is recorded in the history.	Pass
TC45	F239	Delete a non-existent voxel.	System returns an error indicating the voxel does not exist.	Asserts verify that attempting to delete a non-existent voxel leaves the database unchanged.	Pass
TC46	F2310	Reset materials on selected voxels.	The voxels are reset to the default material in the database.	Asserts verify that selected voxels have their material reset to the default value, magnetization remains unchanged, and the reset is recorded in the history.	Pass
TC47	F2310	Reset magnetizations on selected voxels.	The voxels are reset to the default magnetization in the database.	Asserts verify that selected voxels have their magnetization reset to the default value in the database, and the reset is recorded in the history.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC48	F2310	Reset materials and magnetizations on selected voxels.	The voxels are reset to the default material and magnetization in the database.	Asserts verify that both material and magnetization are simultaneously reset to their default values for all selected voxels.	Pass

6.3.2 Utilizing the history manager

Table 11 gives the results for history manager tests (TC49–TC57): undo, redo, and clear history.

Table 11: Editing Manager: Utilizing the history manager (backend)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC49	F237	Undo any updates made to a voxel.	The changes are undone and the voxel returns to the previous state.	Asserts verify that undoing a voxel update restores the previous state in the database, moves the change to the redo stack, and returns a success response.	Pass
TC50	F237	Redo any undone updates to a voxel.	The changes are redone and the voxel returns to the previous state.	Asserts verify that redoing an undone update reapplies the change to the database, clears the redo stack, and returns a success response.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC51	F237	Undo any added voxels.	The voxels are removed from the database.	Asserts verify that undoing an add operation removes the previously added voxels from the database and returns a success response.	Pass
TC52	F237	Redo any deleted voxels.	The voxels are added back to the database.	Asserts verify that redoing a delete operation restores the previously removed voxels to the database and returns a success response.	Pass
TC53	F237	Undo any reset material on a voxel.	The voxel is reset to the default material.	Asserts verify that undoing a reset material operation restores the original voxel data and returns a success response.	Pass
TC54	F237	Redo any reset material on a voxel.	The voxel is reset to the default material.	Asserts verify that redoing a reset material operation re-applies the reset and returns a success response.	Pass
TC55	F237	Undo when the history is empty.	System returns an error indicating the history is empty	Asserts verify that attempting to undo with an empty history raises an error and returns a 400 response.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC56	F237	Redo when the redo stack is empty.	System returns an error indicating the redo stack is empty	Asserts verify that attempting to redo with an empty redo stack raises an error and returns a 400 response.	Pass
TC57	F237	Clear history.	Both history and redo stacks are empty.	Asserts verify that clearing the history empties both the undo and redo stacks, and that recording a new change after an undo also clears the redo stack.	Pass

6.3.3 Saving project

Table 12 gives the results for saving project tests (TC58–TC61).

Table 12: Editing Manager: Saving project (backend)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC58	F236	Save the project automatically.	The project is saved to the database.	Asserts verify that the project is saved to the database when the user makes changes and the auto-save feature is enabled.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC59	F2311	Save the project manually.	The project is saved to the database.	Asserts verify that saving a project triggers a file download and confirms the filename in a success message.	Pass
TC60	F2311	Save a non-existent project.	System returns an error indicating the project does not exist.	Asserts verify that attempting to save without a loaded project displays an appropriate error message via both the menu and the keyboard shortcut.	Pass
TC61	F2311	Save a project with invalid data.	System returns an error indicating the project data is invalid.	Asserts verify that a failed save operation displays an error message to the user.	Pass

6.4 Export Manager Tests

6.4.1 Export Validation and File Export

Table 13 gives the results for export validation and file export (TC62–TC70).

Table 13: Export Manager Test Cases

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC62	F241	Validate all voxels have magnetization.	System checks if all voxels have magnetization and returns an error if any voxels are missing magnetization.	Asserts verify that validation fails when any voxel has both polar and azimuth magnetization angles at their default (unset) value of zero, whether for a single voxel or all voxels in the project.	Pass
TC63	F241	Validate all voxels have material.	System checks if all voxels have material and returns an error if any voxels are missing material.	Asserts verify that validation fails when any voxel has a material ID of zero (unassigned), whether for a single voxel or all voxels in the project, and that validation passes when all voxels have valid materials and magnetization.	Pass
TC64	F242	Export CSV file of an existing project.	System exports a CSV file with all voxel metadata.	Asserts verify that the exported CSV contains the correct column headers and voxel data rows, is returned as a downloadable attachment with the correct filename, and the export function completes successfully.	Pass
TC65	F242	Export CSV file of a non-existent project.	System returns an error indicating the project does not exist.	Asserts verify that attempting to export a non-existent project returns a 404 error.	Pass

Continued on next page

(Continued from previous page)

TC	Ref. Req.	Action	Expected Result	Actual Result	Result
TC66	F242	Export CSV file of a project with invalid data.	System returns an error indicating the project data is invalid.	Asserts verify that exporting a project with incomplete voxel data returns a 400 error, and that the export function returns false for an empty project directory.	Pass
TC67	F243	Export project file of an existing project.	System exports a project file with all project data.	Asserts verify that the project file is exported successfully with the correct column headers, data rows, and downloadable attachment headers.	Pass
TC68	F243	Export project file of a non-existent project.	System returns an error indicating the project does not exist.	Asserts verify that attempting to export a non-existent project file returns a 404 error.	Pass
TC69	F243	Export project file of a project with invalid data.	System returns an error indicating the project data is invalid.	Asserts verify that a database error during export returns a 500 error response.	Pass
TC70	F241	Export operation is interrupted (simulated failure).	The original project data remains intact and unaltered.	Asserts verify that when an export operation fails, any temporary files are cleaned up and the original project data remains intact.	Pass

7 Changes Due to Testing

During the VnV process, several changes to the project that were made based on the feedback from the stakeholders (supervisor and graduate students) that will be used to improve the system. These changes are related to the usability and user input, interaction flow, and performance adjustments.

7.1 Usability and User Input Adjustments

Feedback from usability testing led to the following usability and user-input changes:

- **Project and file menus:** A project popup after selecting a file to show existing projects, improving feedback and selection, as it was not clear where to navigate to open a project.
- **Layer editor:** The magnitude control in the layer editor was removed to reduce clutter and confusion and as it was not used by the stakeholders.
- **File and material naming:** File naming was reviewed to handle spaces and to indicate when a file with the same name already exists. Material naming in settings was reviewed for clarity and consistency.
- **Magnetization display:** A checkmark or toggle was added to show or hide magnetization, giving users control over what is displayed.
- **Partition behaviour:** Default partition centring was changed so partitions are not automatically centred. A rename-partition capability was added.

7.2 Interaction Flow Adjustments

The following changes address the sequence of user actions and overall flow from opening a file to working with partitions:

- **Partition visibility:** The cause of partitions not showing immediately was investigated and addressed so that partitions appear promptly after selecting a project or configuring a new one.
- **Reduced redundancy:** When a file is already open, re-selecting or re-confirming the same file is no longer required for subsequent actions where possible.

7.3 Performance Adjustments

Performance feedback from testing led to optimisations focused on model loading and rendering:

- **Flow from first open to partitions:** The slowness of the flow from first opening a file to obtaining partitions was addressed through performance improvements in file loading and initial processing.

- **Model rendering:** Trimesh (or equivalent mesh handling) was used or integrated to speed up model rendering, reducing latency when displaying the voxelised model and partitions.

7.4 Future Improvements

Usability testing also highlighted potential enhancements for future work:

- **Partition logic:** Revisit or extend partition logic to support more flexible partitioning strategies and user control.
- **Lasso tool:** Consider a lasso (or similar) selection tool to add voxels in a more intuitive way.
- **Export Model Summary:** Add a model summary to the export model file to allow users to see what changes were made to the model.
- **Favourite Magnetization Vectors:** Add a favourite magnetization vectors feature to the system to allow users to save and reuse frequently used magnetization vectors.
- **Export Progress Trackers:** Test the export performance of the system to ensure it meets the requirements and improve the progress tracker.

Overall, the testing phase was successful and played a crucial role in the development of the system. It allowed for the identification of issues and the implementation of improvements, ensuring the system is of high quality and meets the requirements of the stakeholders.

8 Automated Testing

All tests for the Python backend and for the TypeScript frontend modules are automated. The pass/fail results in the tables in Section 6 are backed by the output of these test runs, readers can reproduce the evidence by running the tests as described in the VnV Plan.

- **Backend (Python/PyTest):** From the project root, run `pytest` in `src/backend/` (e.g. `cd src/backend && pytest`). The console output shows per-test pass/fail and any failures; this output is the evidence for the "Actual Result" and "Result" columns in the tables in Section 6.
- **Frontend (TypeScript/Jest):** From `src/frontend/`, run `npm run test`. The console (or test report) output is the evidence for the "Actual Result" and "Result" columns in the tables in Section 6.

System-level and UI tests are executed manually; their outcomes are recorded in the corresponding tables in Section 6 and match the expected results listed there.

9 Trace to Requirements

The test cases in this report are the execution of the system and unit tests defined in the VnV Plan. The following tables show the traceability between these test cases and the functional requirements from the SRS. A filled cell (■) indicates that the test case verifies the corresponding requirement; an empty cell (□) indicates it does not.

9.1 Functional Requirements

Table 14 shows the trace between Import Manager test cases and corresponding requirements.

Table 14: Trace Between Import Manager Test Cases and Corresponding Requirements

Reqs:	F211	F212	F213	F214	F215	F216
TC1	■	□	□	□	□	□
TC2	■	□	□	□	□	□
TC3	■	□	□	□	□	□
TC4	■	□	□	□	□	□
TC5	□	■	□	□	□	□
TC6	□	□	■	□	□	□
TC7	□	□	□	■	□	□
TC8	□	□	□	□	□	■
TC9	□	□	□	□	□	■
TC10	□	□	□	□	□	■
TC11	□	□	■	■	■	□
TC12	□	□	■	■	■	□

Table 15 shows the trace between Visualization Manager test cases and corresponding requirements.

Table 15: Trace Between Visualization Manager test cases and corresponding requirements

Reqs:	F221	F222	F223	F224	F225	F226	F227
TC13	■	□	□	□	□	□	□
TC14	■	□	□	□	□	□	□
TC15	□	■	□	□	□	□	□
TC16	□	■	□	□	□	□	□
TC17	□	■	□	□	□	□	□
TC18	□	□	■	□	□	□	□
TC19	□	□	■	□	□	□	□
TC20	□	□	□	■	□	□	□
TC21	□	□	□	□	■	□	□
TC22	□	□	□	□	■	□	□
TC23	□	□	□	□	■	□	□
TC24	□	□	□	□	■	□	□
TC25	□	□	□	□	□	■	□
TC26	□	□	□	□	□	□	■

Table 16 shows the trace between Editing Manager test cases and corresponding requirements (Part 1: voxel editing and history).

Table 16: Trace Between Editing Manager Test Cases and Corresponding Requirements (Part 1: voxel editing and history)

Reqs:	F231	F232	F233	F234	F235	F237
TC27	■	□	□	□	□	□
TC28	■	□	□	□	□	□
TC29	■	□	□	□	□	□
TC30	■	□	□	□	□	□
TC31	□	□	■	□	□	□
TC32	□	□	■	□	□	□
TC33	□	□	■	□	□	□
TC34	■	□	■	□	□	□
TC35	□	□	□	■	□	□
TC36	□	□	□	□	■	□
TC37	□	□	□	□	□	■
TC38	□	□	□	□	□	■
TC39	□	□	□	□	□	■
TC40	□	□	□	□	□	■
TC41	□	□	□	□	□	■
TC42	□	□	□	□	□	■
TC43	□	□	□	□	□	■
TC44	□	□	□	□	□	■
TC45	□	□	□	□	□	■

Table 17 shows the trace between Editing Manager test cases and corresponding requirements (Part 2: layers, add/delete, reset, save).

Table 17: Trace Between Editing Manager Test Cases and Corresponding Requirements (Part 2: layers, add/delete, reset, save)

Reqs:	F238	F239	F2310	F2311
TC46	■	□	□	□
TC47	■	□	□	□
TC48	■	□	□	□
TC49	■	□	□	□
TC50	■	□	□	□
TC51	■	□	□	□
TC52	□	■	□	□
TC53	□	■	□	□
TC54	□	■	□	□
TC55	□	□	■	□
TC56	□	□	■	□
TC57	□	□	■	□
TC58	□	□	□	■
TC59	□	□	□	■
TC60	□	□	□	■
TC61	□	□	□	■

Table 18 shows the trace between Export Manager test cases and corresponding requirements.

Table 18: Trace Between Export Manager Test Cases and Corresponding Requirements

Reqs:	F241	F242	F243	F244	F245
TC62	■	□	□	□	□
TC63	■	□	□	□	□
TC64	□	■	□	□	□
TC65	□	■	□	□	□
TC66	□	■	□	□	□
TC67	□	□	■	□	□
TC68	□	□	■	□	□
TC69	□	□	■	□	□
TC70	■	□	□	□	□

9.2 Non-Functional Requirements

Table 19 shows the trace between Import Manager Non-Functional Tests and corresponding requirements.

Table 19: Trace Between Import Manager Non-Functional Tests and Corresponding Requirements

Reqs:	NF211	NF212	NF213
TC8	■	□	□
TC9	■	□	□
PS-1 (Performance Test)	■	□	□
PS-2 (Performance Test)	■	■	□
US-3 (Usability Test)	□	□	■

Table 20 shows the trace between Visualization Manager Non-Functional Tests and cor-

responding requirements.

Table 20: Trace Between Visualization Manager Non-Functional Tests and Corresponding Requirements

Reqs:	NF221	NF222	NF223
TC18	■	□	□
TC19	■	□	□
TC15	□	■	□
TC16	□	■	□
PS-1 (Performance Test)	□	■	□
PS-2 (Performance Test)	■	□	□
US-3 (Usability Test)	□	□	■

Table 21 shows the trace between Editing Manager Non-Functional Tests and corresponding requirements.

Table 21: Trace Between Editing Manager Non-Functional Tests and Corresponding Requirements

Reqs:	NF231	NF232
TC59	■	□
US-3 (Usability Test)	■	□
PS-2 (Performance Test)	□	■

Table 22 shows the trace between Export Manager Non-Functional Tests and corresponding requirements.

Table 22: Trace Between Export Manager Non-Functional Tests and Corresponding Requirements

Reqs:	NF241	NF242	NF243
TC63	■	■	□
TC64	■	■	□
TC65	■	■	□
TC66	■	■	□
TC67	■	■	□
CO-3 (Portability Test)	■	□	□
US-2 (Usability Test)	□	■	□
PS-2 (Performance Test)	□	■	□
Not Explicitly Tested	□	□	■

Table 23 shows the trace between General Non-Functional Tests and corresponding requirements.

Table 23: Trace Between General Non-Functional Tests and Corresponding Requirements

Reqs:	GEN1	GEN2
US-6 (Usability Test)	■	□
MA-1 (Maintainability Test)	□	■

10 Trace to Modules

The following tables show the traceability between the test cases in this report and the modules defined in the Module Guide. A filled cell (■) indicates that the test case exercises or verifies behaviour involving that module; an empty cell (□) indicates it does not. The module identifiers correspond to the modules in the MG.

Table 24 shows the trace between Import Manager test cases and modules.

Table 24: Trace Between Import Manager Test Cases and Modules

TC	M1	M2	M3	M4	M5
TC1	■	□	□	■	□
TC2	■	□	□	■	□
TC3	■	□	□	■	□
TC4	■	□	□	■	□
TC5	■	□	□	■	□
TC6	■	■	□	□	■
TC7	■	□	□	□	■
TC8	□	□	■	□	□
TC9	□	□	■	□	□
TC10	□	□	■	□	□
TC11	■	■	□	■	□
TC12	■	■	□	■	□

Table 25 shows the trace between Visualization Manager test cases and modules.

Table 25: Trace Between Visualization Manager Test Cases and Modules

TC	M3	M5	M6	M9	M10	M14	M15
TC13	☐	☐	■	☐	☐	☐	☐
TC14	☐	☐	■	☐	☐	☐	☐
TC15	■	■	■	☐	☐	■	☐
TC16	■	■	■	☐	☐	■	☐
TC17	■	■	■	☐	☐	■	☐
TC18	☐	■	■	☐	☐	■	■
TC19	☐	■	■	☐	☐	■	■
TC20	☐	■	■	☐	☐	■	■
TC21	☐	■	■	■	■	■	■
TC22	☐	■	■	■	■	■	■
TC23	☐	■	■	■	■	■	■
TC24	☐	■	■	■	■	■	■
TC25	☐	☐	■	■	■	■	■
TC26	☐	☐	■	■	■	■	■

Table 26 shows the trace between Editing Manager test cases and modules (Part 1: voxel editing and history).

Table 26: Trace Between Editing Manager Test Cases and Modules (Part 1: voxel editing and history)

TC	M5	M7	M8	M9	M10	M13	M14
TC27	■	■	□	□	□	□	■
TC28	■	■	□	□	□	□	■
TC29	■	■	□	□	□	□	■
TC30	■	■	□	□	□	□	■
TC31	■	■	□	□	□	□	■
TC32	■	■	□	□	□	□	■
TC33	■	■	□	□	□	□	■
TC34	■	■	□	□	□	□	■
TC35	■	■	□	□	■	□	□
TC36	■	■	■	■	□	□	■
TC37	■	■	■	□	□	□	□
TC38	■	■	■	□	□	□	□
TC39	■	■	■	□	□	□	□
TC40	■	■	■	□	□	□	□
TC41	■	■	■	□	□	□	□
TC42	■	■	■	□	□	□	□
TC43	■	■	■	□	□	□	□
TC44	■	■	■	□	□	□	□
TC45	■	■	■	□	□	□	□

Table 27 shows the trace between Editing Manager test cases and modules (Part 2: layers, add/delete, reset, save).

Table 27: Trace Between Editing Manager Test Cases and Modules (Part 2: layers, add/delete, reset, save)

TC	M4	M5	M7	M10	M14
TC46	☐	■	■	■	■
TC47	☐	■	■	■	■
TC48	☐	■	■	■	■
TC49	☐	■	■	■	■
TC50	☐	■	■	■	■
TC51	☐	■	■	■	■
TC52	☐	■	■	☐	■
TC53	☐	■	■	☐	■
TC54	☐	■	■	☐	■
TC55	☐	■	■	☐	■
TC56	☐	■	■	☐	■
TC57	☐	■	■	☐	■
TC58	■	☐	☐	☐	☐
TC59	■	☐	☐	☐	☐
TC60	■	☐	☐	☐	☐
TC61	■	☐	☐	☐	☐

Table 28 shows the trace between Export Manager test cases and modules.

Table 28: Trace Between Export Manager Test Cases and Modules

TC	M9	M11	M12	M13	M14	M16
TC62	■	■	□	□	■	□
TC63	■	■	□	□	■	□
TC64	□	□	■	□	■	■
TC65	□	□	■	□	■	■
TC66	□	□	■	□	■	■
TC67	□	□	■	□	■	□
TC68	□	□	■	□	■	□
TC69	□	□	■	□	■	□
TC70	■	■	□	□	■	□

11 Code Coverage Metrics

11.1 Python Code Coverage

The following section analyzes the code coverage of the Python code in the backend. The analysis is provided in the Figure 4 and is a measure of how much of the code is being tested, and helps us understand which parts of the code are not being tested.

Coverage report: 90%

Files | Functions | Classes

filter...

hide covered

coverage.py v7.13.4, created at 2026-03-08 20:10 -0400

File	statements	missing	excluded	coverage
app/__init__.py	0	0	0	100%
app/config.py	15	1	0	93%
app/models/__init__.py	2	0	0	100%
app/models/schemas.py	44	0	0	100%
app/routers/__init__.py	0	0	0	100%
app/routers/edit_router.py	144	11	0	92%
app/routers/export_router.py	29	0	0	100%
app/routers/project_router.py	70	27	0	61%
app/routers/stl_router.py	41	7	0	83%
app/services/__init__.py	0	0	0	100%
app/services/export_service.py	21	0	0	100%
app/services/history_management_service.py	39	4	0	90%
app/services/mesh_service.py	6	3	0	50%
app/services/model_editing_service.py	38	0	0	100%
app/services/model_structure_service.py	98	6	0	94%
app/services/model_tracking_service.py	51	0	0	100%
app/services/partition_manager.py	39	0	0	100%
app/services/project_management_service.py	36	5	0	86%
app/services/voxel_service.py	7	3	0	57%
Total	**680**	**67**	**0**	**90%**

coverage.py v7.13.4, created at 2026-03-08 20:10 -0400

Figure 4: Python Code Coverage

The backend codebase has an overall code coverage of 90% for statements, branches, functions, and lines. The code coverage overall met the VnV Plan's requirement of 50%. The `voxel_service.py` was the file that was the most difficult to test because it utilizes the `trimesh` library for voxelization. The team will furthur analyze the code coverage of the `voxel_service.py` file to understand which parts of the code are not being tested and improve the coverage of the file.

11.2 TypeScript Code Coverage

The following section analyzes the code coverage of the TypeScript code in the frontend. The analysis is provided in the Figure 5 and is a measure of how much of the code is being tested, and helps us understand which parts of the code are not being tested.

```
-----------------------|---------|----------|---------|---------|-------------------------------
File                   | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------------------|---------|----------|---------|---------|-------------------------------
All files              |   83.08 |    65.98 |   85.39 |   84.14 |
 src                   |   84.41 |       50 |   79.36 |   85.32 |
  App.tsx              |   84.41 |       50 |   79.36 |   85.32 | 113,132,157-163,169-170,195-197,
 src/components        |    72.9 |    66.74 |   88.18 |   73.47 |
  Footer.tsx           |     100 |      100 |     100 |     100 |
  LayerEditor.tsx      |   58.56 |     46.8 |   76.59 |   59.19 | 111-114,133-136,146-148,155-167,
  MenuBar.tsx          |     100 |    96.77 |     100 |     100 | 116,290,338
  ModelSelector.tsx    |     100 |      100 |     100 |     100 |
  NewProjectDialog.tsx |   86.31 |    78.26 |   90.47 |   88.37 | 91-102,238
  PartitionsPanel.tsx  |     100 |    94.11 |     100 |     100 | 33
  ProjectSelector.tsx  |     100 |      100 |     100 |     100 |
  StatusMessage.tsx    |     100 |      100 |     100 |     100 |
  UploadButton.tsx     |      75 |     62.5 |     100 |   72.72 | 12,16-17
  UploadMessage.tsx    |     100 |      100 |     100 |     100 |
 src/utils             |   97.04 |    78.81 |   86.79 |   99.69 |
  api.ts               |   96.37 |    80.32 |   81.08 |     100 | 38,53-58,106,157,173,218,326-327
  threeUtils.ts        |   97.76 |    77.19 |     100 |   99.37 | 293
-----------------------|---------|----------|---------|---------|-------------------------------

============================== Coverage summary ==============================
Statements   : 83.08% ( 1022/1230 )
Branches     : 65.98% ( 452/685 )
Functions    : 85.39% ( 193/226 )
Lines        : 84.14% ( 950/1129 )
==============================================================================

Test Suites: 16 passed, 16 total
Tests:       221 passed, 221 total
Snapshots:   0 total
Time:        4.317 s, estimated 6 s
Ran all test suites.
```

Figure 5: TypeScript Code Coverage

The frontend codebase has an overall code coverage of 83.08% for statements, 65.98% for branches, 85.39% for functions, and 84.14% for lines. The code coverage overall met the VnV Plan's requirement of 50%. The team will continue to improve the coverage of the frontend codebase in the future.

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

Omar: The main thing that went well was connecting the VnV plan with the VnV report. The plan already defined the tests clearly, so it was straightforward to structure the report around usability, performance, portability, maintainability, and security. The usability testing also went well since we were able to get feedback from both technical and non-technical users, which helped validate that the interface and main features were intuitive.

Daniel: I worked on the backend unit tests, as well as the organization of the usability test with our supervisor and her graduate students, which both went fairly smoothly in spite of the time crunch. The reorganization of the backend into cohesive modules and functions that I participated in was extremely helpful when generating all unit test cases, as well as their relative simplicity. The structure of the usability test - providing as little instruction as possible - also made it easy to identify parts of the software that still need work in terms of understandability.

Andrew: I worked on UI unit tests, listing all functional requirements in the report, and explaining the alternative COMSOL to AutoVox. Writing the UI tests went well, and I learned a lot about mocking modules, and making sure functions that needed to be fired and validated helped find any errors in the codebase.

Olivia: During this deliverable, I helped write some of the backend unit tests and was the primary person writing the usability report after usability testing, which was conducted by all group members. Due to the nature of this deliverable, I think that it really helped us narrow down some of the major areas that require improvement and further development as well as build confidence in the code that was already developed. With the final deadlines quickly approaching, there is a lot to do in very little time. Therefore, this type of direction is important to ensure that we make the best use out of the time we have remaining.

Khalid: I worked on the unit tests and traceability between the test cases and the requirements. Having the traceability between the functional requirements and modules in the MIS document was very helpful in ensuring that all the test cases were covered and that the test cases were testing the correct functionality. It was also helpful to have the traceability between the test cases and the modules in the VnV Plan document. Overall, this deliverable went really well as everything was clearly defined and we were able to complete it on time. Another thing that was helpful was the fact that I didn't have much university work to do at the same time, so I was able to focus on this deliverable and get it done on time.

2. What pain points did you experience during this deliverable, and how did you resolve them?

Omar: One difficulty was making sure the report matched the VnV plan without repeating the same information. Since the plan already explains how tests are performed, the report had to focus more on results and evaluation. Another challenge was presenting the nonfunctional testing clearly, especially performance results, which required organizing the data into tables and graphs. Following the structure of the reference VnV report helped resolve this.

Daniel: Less of a pain point and more a frustration due to lack of experience; when generating unit tests my primary struggle was learning proper use of pytest and mocking classes, as well as when to mock. While I had planned to become familiar with it prior to needing to write these tests, other commitments prevented me from doing so, and I had to learn on the fly how to accomplish what I wanted to do. Fortunately, it was less complex once I figured out the syntax and I was able to finish all unit tests by the deadline I promised.

Andrew: I didn't experience many pain points, however it was hard to see exactly what our report should look like. The rubric wasn't that detailed, and the only way to see what to do correctly was to check prior submissions on the GitLab.

Olivia: In relation to the usability testing, one of the challenges that we experienced was choosing exactly how much guidance to give the user as they interacted with the software. We want to be able to provide enough structure and context so that the user is able to understand what we want them to do and how they can do it, while not making instructions too explicit in a way that compromises how a user would intuitively interact with and navigate through the software. While we went into the usability testing with a plan, things were definitely adapted in the moment. I think this is a natural part of usability testing. While you can have an idea on how you think people will interact with something, reality isn't always what you imagine. If we decide to conduct another round of usability testing within the coming weeks after further developing the software, there will definitely be changes in the structure and method we choose to conduct usability testing that reflect some of our observations during the first round.

Khalid: I didn't experience much pain points during this deliverable. The only pain point I had was just the amount of repetitive work that was required to document the test cases and the requirements. However, I was able to resolve it by breaking down the work into smaller tasks and completing them one by one.

3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?

Several parts of this document came from discussions with our supervisor and with a proxy (our peers). The scope of **functional and non-functional testing**—what needed to be tested and how, was conducted by speaking to the proxy and jointly defining the test objectives and coverage. The **Comparison to Existing Implementation** section came from speaking to our supervisor and to the graduate students who use the lab's 3D printing workflow. Those conversations helped us understand how they currently complete their objectives (e.g., with COMSOL) and what AutoVox should improve or replace. **Unit testing**, **automated testing**, and **coverage** were driven mainly by the proxy, we relied on peer discussion and shared standards to decide frameworks (e.g., PyTest, Jest), what to automate, and what coverage targets to aim for. **Changes due to testing** came from a mix of both: the proxy helped prioritize and validate technical changes, while the supervisor and graduate students provided context on whether those changes aligned with real usage and research needs.

4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)

We did not make any major changes to the VnV Plan, the activities we conducted aligned with what was planned. We were able to predict a feasible amount of effort and the right tasks mainly because the plan was well scoped and supported by strong traceability. The traceability between the functional requirements (SRS), the modules (MIS and Module Guide), and the test cases in the VnV Plan made it clear which tests were needed and what they had to cover, so we could estimate effort and avoid gaps or duplicate work. The backend had also been reorganized into cohesive modules and functions before we wrote the unit tests, which made the test cases straightforward to implement and kept the workload manageable. For usability testing, the plan specified the overall approach (e.g., exploratory testing with minimal instruction and a task-based evaluation) but left out the exact list of tasks. When we ran the usability test, we did refine the specific tasks and the level of guidance given to participants (e.g. how much context to provide before each task). These were minor adjustments rather than changes to the plan itself, and they did not affect the planned objectives or deliverables. In future projects, we would again rely on clear traceability and a plan that

fixes the strategy and scope while allowing small operational details (like concrete usability tasks) to be finalized at execution time.